Predicting Zero-Shot Video-Tracker Transfer: Methods, Results, and Engineering Log

Progress report — "will-it-track"

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Abstract

We are testing whether the zero-shot tracking performance of a frozen promptable video model (SAM 3) on an *unseen* species or place can be predicted *before running it*, from four label-free "distances" (taxonomic, visual, environment, temporal), and whether the two halves of tracking — *finding* the animal (pDetA) and *following* it (pAssA) — are governed by different factors. This document records, in plain language, exactly what has been built and run so far, what the results say, and every practical and statistical problem we hit together with the fix we applied. The headline so far is an *honest null*: on the two splits tested, the distances do not predict SAM 3's per-cell detection/association to a degree any valid test supports. This is a legitimate scientific result, and it points clearly at the next experiment (a powered species hold-out), which is running now.

1 Aim and hypotheses

Conservation teams point promptable trackers at new species and new camera-trap sites constantly, but nobody can say in advance whether the model is trustworthy for a given animal in a given place. We try to fill that gap with a *reliability estimator*: given only label-free distances, predict pDetA and pAssA with honest confidence intervals. The guiding hypotheses are:

H1 (detection). pDetA falls with **species novelty** (taxonomic + visual distance).

H2 (association). pAssA falls with **environment difficulty** (scene distance, day/night/IR, clutter), largely regardless of species.

H0 (null). Distance explains little variance — *still a valid result*, and one that would redirect the project toward representational probing.

The goal is explicitly **not** to make the tracker score higher; it is to *predict* and *explain* its transfer.

2 Data

We use **SA-FARI** (Meta × Conservation X Labs, 2025), the largest open multi-animal tracking dataset: 11,609 camera-trap videos, 99 species with a full 7-level taxonomy, 741 locations on 4 continents, and per-video timestamps. Annotations are on 6 fps frames (mean ≈85 frames per clip). Crucially, SAM 3 is **frozen and zero-shot** on all of SA-FARI, so the train/test split is only our own reference anchor for measuring distances — not anything the model learned.

3 Method

The pipeline has five stages; each is a small, config-driven module.

3.1 Inference

A frozen SAM 3 promptable tracker is run over every probe of a split. For each video we give a text prompt (the species name, e.g. `"impala"`) and the model segments and tracks every matching animal across the clip. Predictions are written one JSON per video, so the run is *resumable*. Hard negatives (a query for an absent species) are kept — they should return nothing.

3.2 Scoring

We score with the *official* SA-Co / VEval evaluator (never re-implemented). The promptable HOTA score decomposes cleanly into the two parts we analyse separately:

$$\mathsf{pHOTA} = \sqrt{\mathsf{pDetA} \times \mathsf{pAssA}}, \qquad \mathsf{pDetA} = \text{did it find the animals?}, \quad \mathsf{pAssA} = \text{did it keep each identity over}$$

Scores are recorded per **cell** — a (species, location, time) group — together with the *support* behind each score (number of frames, masklets, videos).

3.3 The four label-free distances

Every distance is computable *without any label of the target species*, which is what makes a before-running prediction possible.

Distance	What it measures	Granularity
Taxonomic	Tree distance (Kingdom→Species) to the nearest reference species	per species
Visual	$1 - \cos$ between DINOv2 embeddings of the *mask-cropped animal* and the nearest reference species' prototype	per species
Environment	$1 - \cos$ between DINOv2 embeddings of the *masked-background scene* and the nearest reference location's prototype; plus a night/IR flag and a clutter score	per location
Temporal	Gap (in years) between the probe footage and the reference footage	per cell

Table 1: The four distances. Visual and environment reuse a shared, cache-first embedding pipeline.

3.4 The two experiments (splits)

Split B — location hold-out (secondary, tests H2). Reference = official train split, probe = test split. Camera sites are unseen; species are largely shared, so the taxonomic distance is ≈ 0 and the environment/temporal distances vary. This is the "does the scene matter?" experiment.

Split A — species hold-out (primary, tests H1). Leave-*one*-species-out over the pooled present set: each probe species' distance is to the nearest *other* species, which makes taxonomic novelty a genuine *graded* axis. This is the "does novelty matter?" experiment.

3.5 Modelling

For each target (pDetA, pAssA) we fit a support-weighted logit-link GLM (a fractional-logit model on the bounded score), with the four distances plus the scene covariates and a log(support) covariate so that "rare" is never mistaken for "far". Predictors are standardised. We then:

- report **group-cluster-bootstrap** confidence intervals (refit while resampling whole species / locations) — see the confidence-interval issue below;

- run **grouped cross-validation** (leave-one-species-out and leave-one-location-out) and test the out-of-sample gain over a mean predictor with a **paired group-bootstrap** *p*-value;
- partition variance across factors (dominance analysis) and check collinearity (VIF).

4 Results so far

4.1 Gate 1 — measurement (is SAM 3 behaving sensibly?)

Zero-shot SAM 3 on the SA-FARI test split reproduces the published ballpark. The dataset-level (micro) mask pHOTA is ≈ 0.65 with a real detection–association gap. Table 2 reports the per-cell (macro) support-weighted means we feed to the models; these are lower than the micro number because 78% of cells are single-object clips where pAssA≈pDetA (finding and following coincide when there is only one animal).

Metric (mask, per-cell macro mean)	Value
pDetA	0.527
pAssA	0.546
pHOTA	0.533
Cells scored	346 of 1447

Table 2: Gate 1 measurement on the test split. The 1101 unscored cells are hard negatives (species absent, no ground-truth track), correctly excluded from the HOTA fit and reserved for a separate false-positive analysis.

4.2 Gate 2 — location split (Split B): a null

After correcting the inference (the confidence-interval issue below), **every distance coefficient's confidence interval spans zero**, and the out-of-sample gain over a mean predictor is **not significant** on either cross-validation scheme (Tables 3–5). The large-looking taxonomic coefficient (-0.36) is a fragile artifact of this split (see the novelty-axis issue below), not a graded effect.

Factor	pDetA coef. [95% CI]	pAssA coef. [95% CI]
Taxonomic distance	−0.362 [−3.87, 0.09]	−0.368 [−3.40, 0.08]
Visual distance	−0.144 [−0.50, 0.14]	−0.122 [−0.49, 0.18]
Environment distance	+0.042 [−0.19, 0.24]	+0.013 [−0.23, 0.21]
Clutter	−0.066 [−0.38, 0.29]	−0.087 [−0.41, 0.28]
Night / IR	−0.405 [−0.94, 0.11]	−0.462 [−1.02, 0.11]
log(support)	−0.216 [−0.51, 0.17]	−0.224 [−0.53, 0.18]

Table 3: Split B (location) standardised coefficients with group-cluster-bootstrap CIs. Every distance spans zero. The temporal gap is omitted: it is constant on this split and carries no information.

4.3 Gate 2 — species hold-out (Split A, preview): a cleaner null

Because the location split cannot properly test novelty, we ran the species hold-out (leave-one-species-out) on the already-scored test species — a free preview that needs no new inference. With a *graded* novelty axis, the taxonomic effect **collapses to essentially zero** (Table 4), confirming that the location split's -0.36 was an artifact. The out-of-sample gain is again not significant — indeed slightly *negative* (the distance model is no better than, or marginally worse than, predicting the mean).

Factor	pDetA coef. [95% CI]	pAssA coef. [95% CI]
Taxonomic distance	+0.026 [−0.31, 0.31]	+0.008 [−0.33, 0.31]
Visual distance	+0.120 [−0.23, 0.38]	+0.135 [−0.23, 0.40]
Environment distance	−0.025 [−0.30, 0.21]	−0.046 [−0.33, 0.19]
Clutter	−0.106 [−0.37, 0.21]	−0.135 [−0.41, 0.20]
Night / IR	−0.394 [−0.99, 0.09]	−0.466 [−1.09, 0.06]
log(support)	−0.188 [−0.49, 0.19]	−0.193 [−0.51, 0.21]

Table 4: Split A (species hold-out, test-species pool) standardised coefficients. With a graded novelty axis the taxonomic and visual effects are ≈ 0 and span zero. pseudo-R^2 is 0.033 / 0.039.

Split	Hold-out	Target	ΔMAE	95% CI	p
Location (B)	location	pAssA	+0.007	[−0.005, 0.018]	0.13
Location (B)	location	pDetA	+0.004	[−0.007, 0.015]	0.23
Species (A)	species	pAssA	−0.003	[−0.013, 0.006]	0.74
Species (A)	species	pDetA	−0.004	[−0.013, 0.005]	0.79

Table 5: Out-of-sample gain over a mean predictor (ΔMAE > 0 means the model is better), with a paired group-bootstrap CI and one-sided p. **No row is significant** on either split.

4.4 What the results mean (honestly)

On both splits tested, the label-free distances do **not** predict SAM 3's per-cell detection or association to a degree any valid, group-clustered test supports. This is **H0-leaning**. It is not yet "H0 confirmed for the whole thesis": the species hold-out preview uses only the 83 test species, and the powered version (adding a sample of train species) is running now. The two *large-looking* effects that survive as point estimates — night/IR and (on Split B) taxonomic — do not survive an honest interval, and the taxonomic one is a known artifact. The one robust qualitative signal is that **night/IR footage lowers both** pDetA and pAssA (day → night: pDetA 0.60 → 0.51, pAssA 0.62 → 0.52), but even this is not individually significant once clustering is respected.

5 The evidence for the null (H0)

H0 is the hypothesis that *distance explains little variance* — that the four label-free distances carry almost no usable information about how well SAM 3 will track a new species in a new place. This is not a fallback we reached for; it is what six independent lines of evidence point to. Below, each line is stated, then explained in plain terms, then linked back to H0.

How to read the numbers (a short primer). A *coefficient* is the slope of a fitted line: how much the score is predicted to change when a distance goes up by one standard deviation. A *95% confidence interval (CI)* is the range of slopes the data are consistent with; if that range *includes zero*, we cannot rule out that the true effect is *nothing*. A *p-value* is the probability of seeing an effect at least this large if the truth were really zero — small p (below 0.05) means "unlikely to be a fluke"; large p means "easily just noise". *Pseudo-R^2* is a 0–1 measure of how much of the variation the model explains (0 = nothing, 1 = everything). *MAE* (mean absolute error) is the average size of the prediction mistakes; the *baseline* is the MAE you get by ignoring the distances and just guessing the overall average score. *Out-of-sample* means the model is judged on species/locations it never saw during fitting — the only fair test of a predictor.

5.1 The six lines of evidence

1. **The models explain almost none of the variation.** The fitted models reach a pseudo-R^2 of only ≈ 0.055 on the location split and ≈ 0.03–0.04 on the species hold-out (Table 4 caption). In words: even *inside* the data it was fitted on, the model captures around 3–6% of what makes one cell's score differ from another's — and most of even that little is carried by the night/IR flag and the support covariate, not by the novelty/environment distances the thesis is about. A predictor that explains 3–6% of the variance is, for practical purposes, explaining nothing.

2. **Every distance's confidence interval includes zero.** After honest inference (resampling whole species/locations — see the confidence-interval issue in the log), not one of the four distances has a coefficient we can distinguish from zero, on either split or either target (Tables 3 and 4). For example the taxonomic slope on the species hold-out is +0.026 with a CI of $[-0.31, +0.31]$: the data are equally consistent with a small positive effect, a small negative effect, and no effect at all. When every interval straddles zero, the data are telling us they cannot pin down any direction — the signature of H0.

3. **Even the raw, un-modelled correlations are tiny.** Before any modelling, the plain correlation between each distance and the score never exceeds $|r| \leq 0.17$, i.e. $r^2 \leq 0.03$ (at most 3% of variance shared). This matters because it rules out the objection "maybe the model is just badly specified": the weakness is already there in the simplest possible look at the data, not introduced by the GLM.

4. **The predictor fails its own out-of-sample test — the decisive one.** The whole point was a *predictor*: distances in, a useful guess of the score out, for species and places never seen. We tested exactly that with grouped cross-validation (hold out a whole species or a whole location, fit on the rest, predict the held-out group). The distance model does **not** significantly beat the trivial "just guess the average" baseline (Table 5): on the location split the gain is a mere 0.004–0.007 MAE with $p = 0.13$–0.23 (not significant), and on the species hold-out the gain is actually *negative* ($p = 0.74$–0.79) — the distances make the out-of-sample guess slightly *worse* than ignoring them. A predictor that cannot beat "guess the mean" has no practical value, and this is the single strongest piece of evidence because it is the deliverable failing its actual job.

5. **The one strong-looking effect was an artifact that vanished under a cleaner design.** On the location split the taxonomic distance looked important (-0.36). But that split makes taxonomic novelty degenerate: almost all test species also occur in the reference set, so their distance is ≈ 0, and only $\sim$7 genuinely-unseen species sit above zero — the "-0.36" was a fragile contrast driven by those few. When we re-ran with a proper *graded* novelty axis (the species hold-out, where each species is compared to the nearest *other* species), the taxonomic effect collapsed to +0.03/+0.01 (Table 4). An apparent signal that evaporates the moment the design is fixed is exactly what you expect when there is no real effect.

6. **The null is consistent across two independent designs and both targets.** We now have four independent looks — {location split, species split} $\times$ {detection, association} — and all four give the same answer: no significant distance effect, no out-of-sample gain. A genuine effect would tend to surface in at least one of them; four consistent nulls, from

two differently-constructed splits, make a coincidental miss unlikely and strengthen the H0 reading.

5.2 What this does, and does not, establish

H0 here is a *scoped* claim, and it is important to say exactly how far it reaches.

- **It does establish:** these four label-free distances, at the *per-cell* granularity, on the two splits measured, do not predict SAM 3's detection or association to any degree a valid test supports.
- **It does not establish** that *no* label-free signal exists anywhere. Three things are untested and could still carry signal: (i) coarser aggregation (per-species or per-location averages, which are far less noisy than single short clips); (ii) a *familiarity proxy* (separability in SAM 3's own feature space, rather than in DINOv2's); and (iii) the false-positive / hallucination regime — whether a novel-species prompt fires on an *absent* species — which our present analysis, conditioned on the animal being present, never touches.
- **A residual qualitative pattern remains:** night/IR footage lowers both scores (day $\rightarrow$ night: pDetA 0.60 $\rightarrow$ 0.51, pAssA 0.62 $\rightarrow$ 0.52). Environment therefore does matter a little — but through a simple day/night flag, not through the continuous scene-distance the hypothesis proposed, and even this is not individually significant once clustering is respected.

5.3 Why a null is still a result — and where it points

A validated negative result on a genuinely new problem (the first before-running transfer predictor for a promptable video *tracker*, split into detection vs association) is itself a contribution: it tells the field that the "obvious" label-free distances are not enough, and it does so with honest, cluster-aware statistics rather than an over-confident claim. It also *redirects* the project. If transfer is not simply a function of how *far* a new species/scene is, the sharper question becomes whether SAM 3 *internally represents* the structure at all — **representational probing**: instead of "does distance predict the score?", ask "do SAM 3's own features linearly encode taxonomy, novelty, or scene type?". That is a mechanism question, it reuses the same frozen model and data, and it is exactly the pivot H0 was written to motivate.

6 Issues encountered and how we solved them

This section is the engineering and statistics log. Each row is a real problem we hit and the fix applied.

6.1 Engineering / infrastructure

6.2 Statistical / scientific

7 Current status and next steps

- **Done and validated:** the full measurement + modelling pipeline; the location split (Split B, null); the species hold-out preview (Split A on test species, cleaner null); honest clustered inference throughout.
- **Running now:** the *powered* species hold-out — a capped ($\sim$30 videos/species) SAM 3 pass over the train split ($\sim$1500 probes, $\sim$6 h), which will add species and cells to tighten the H1 test.

Issue	Solution applied
Feature assembly ran out of memory at full sampling caps (all cropped frames held in RAM at once on a ~58 GB container).	Rewrote the embedding step to work in *chunks*: load → embed → cache → free, so peak RAM is bounded by one chunk regardless of the caps. RSS stayed at ~7 GB.
Embedding was I/O-bound at ~5 crops/s (the GPU sat idle while single-threaded frame reads from a network filesystem blocked).	Parallelised frame load/crop with a thread pool (~32 workers); the GPU is now the bottleneck.
The official scorer failed on a fresh pod (its import-time dependencies were missing).	Hardened the pod setup to clone the scorer and install its undeclared deps (`iopath, einops, timm, ftfy, regex`).
PyTorch pulled a CUDA build newer than the pod's driver; and `pip` was OOM-killed installing torch (the temp dir was RAM-backed).	Pin torch to the driver's CUDA; move `TMPDIR` onto persistent disk and install with `--no-cache-dir`.
The full train-split inference is ~31,000 probes ≈ **10 days** of GPU time — far too expensive for one experiment.	Added a *per-species cap* (a stratified sample of ~30 positive videos per species, ≈1500 probes ≈6 h) that still gives a graded novelty axis.
A single pathological clip crashed SAM 3's forward pass (a likely CUDA out-of-memory on a long video) and killed the entire multi-hour batch.	Wrapped per-video inference in a guard: a crashing clip is *skipped* (scored as a miss), the GPU cache is freed, and the batch continues.
Long clips could exhaust GPU memory during video preprocessing.	Keep SAM 3 video preprocessing on the CPU (raw storage only), running the model itself on the GPU.
The proxy SSH to the pod allows no `scp` and needs a pseudo-terminal.	Transfer files by `tar`+`base64` over the terminal and decode locally; run the analysis parquets locally where possible.

- **Then, in priority order:** (i) refit Split A with the pooled train+test species; (ii) a multi-object-only association analysis; (iii) the hard-negative false-positive analysis; (iv) the familiarity proxy; (v) if the powered test confirms the null, pivot the thesis toward *representational probing* (does SAM 3 encode taxonomy?), which H0 explicitly motivates.

Bottom line. The engineering works end-to-end and the statistics are now honest. The current, carefully-tested answer is that simple label-free distances do not predict this tracker's zero-shot transfer on the splits measured so far — a clean, publishable negative result that sharpens the next question rather than a failure.

Issue	Solution applied (or proposed)
Over-precise confidence intervals. Weighting each cell by its frame count inflated the effective sample $\sim 85\times$ and shrank every standard error $\sim 9\times$; on top of that, predictors are constant within a species/location (pseudo-replication), so 346 cells are really only $\sim$60–92 independent groups. The naive intervals looked "significant" when they were not.	Replaced the naive intervals with a **group-cluster-bootstrap**: resample whole species / locations, refit, take percentile intervals. Every distance CI then spans zero. *Next:* also report cluster-robust (sandwich) SEs as a cross-check.
No significance test on the cross-validation gain. A +0.005 MAE edge over the baseline was being read as "beats baseline" with no test.	Added a **paired group-bootstrap** on the CV gain (resample whole held-out groups) $\rightarrow$ a CI and one-sided p. All gains are non-significant.
Degenerate temporal feature. On the location split the temporal gap is present for only 32/346 cells and constant among them — an all-zero column after standardising.	Drop zero-variance predictors automatically. Temporal distance is therefore *untested* (not falsified) on this split; it needs a time-based split to test.
Novelty axis was degenerate on Split B. With the train split as reference, the taxonomic distance is ≈ 0 for shared species and > 0 for only $\sim$7 truly-unseen ones, so the "-0.36" coefficient was a fragile 7-species spike-at-zero, not a graded effect.	Run **Split A** (leave-one-species-out), where novelty is graded. The taxonomic effect then collapses to ≈ 0 — confirming the artifact.
Environment sign-flip. The continuous scene distance has a weak *negative* raw correlation but flips to $\approx$ 0/positive in the fit — a suppression symptom of its collinearity with the night/IR flag (both are per-location, from the same background).	Flagged; VIF currently covers only the four distances. *Next:* add night/IR and clutter to the VIF check and refit environment with/without night/IR to show the confound explicitly.
Detection-vs-association is under-identified. 78% of cells are single-object, so $\mathsf{pAssA} \approx \mathsf{pDetA}$ and the two regressions are near-duplicates — the project's core decomposition cannot be tested here.	*Next:* a targeted analysis on the $\sim$22% multi-object cells, where association carries information independent of detection (flagged exploratory, low power).
Hard negatives excluded from the fit. The detection model is fit on animal-present cells only (recall-given-present), so the	*Next:* a separate false-positive-rate-vs-novelty analysis that reintroduces the 1101 hard negatives.